

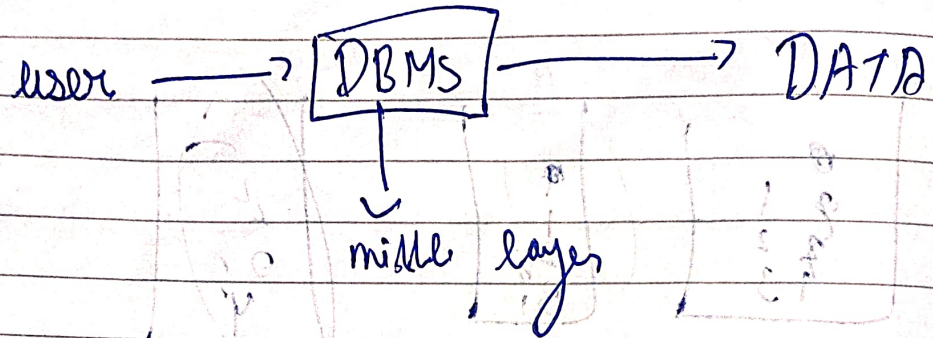
SQL → IBM

classmate

Date
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DATA BASE - collection of DATA

Software that manage DATA is DBMS
(Database management system)



DATA Base

o Relational (RDBMS)

Non relational (NO SQL)

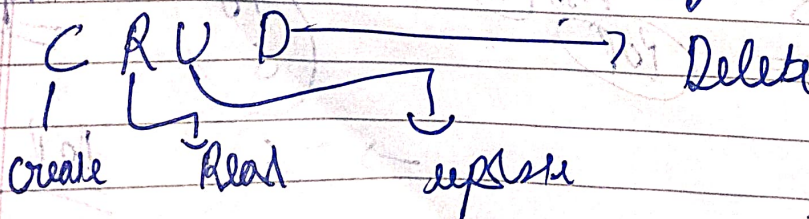
o Table

o No Table

Ex → MySQL, Oracle
SQL server

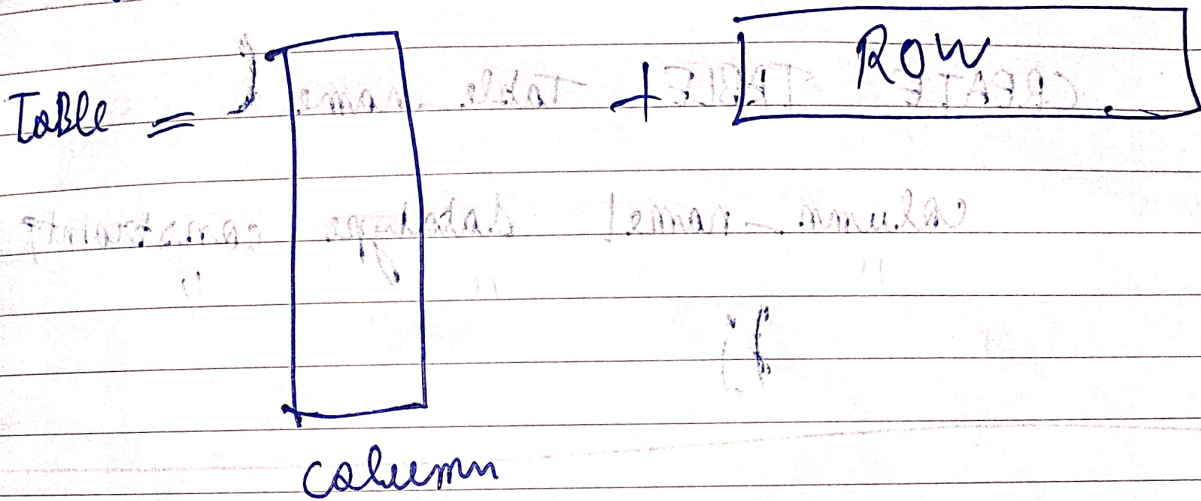
mango DB

SQL → Structured Query language



SQL Structured Query Language

↓
structured
englis
Query
language



create Data base

CREATE DATABASE DB-Name

Delete Data Base

DROP DATABASE DB-Name

use dB-Name → TO select data

create our first Table

USE dB-name

CREATE TABLE Table_name {

column_name | datatype constraint,
" " " "
};

DATA Insert

INSERT INTO dB-NAME VALUES (1, "HARV", 26)

Select

SELECT * FROM student

CHAR Take full memory

VARCHAR Take memory kitini need hai

• BLOB → Big String store (large Binary Object)

• Tiny INT (−128, 127)

• BIG INT

• BIT B(1) → 0, 1

B(2) → {00, 01, 11, 01}

• float → 2.3, 4.56, 456.7

• Double → decimal with 24 to 53 digit

• BOOLEAN → 0 to 1

• DATE: YY-YY-MM-DD

• YEAR — 2001, 2005, 1999, 2022

+ - $\rightarrow$ signed
+ $\rightarrow$ unsigned

~~TINYINT~~ TINYINT UNSIGNED (0 to 255)

TINYINT (-128 to 127)

Types of SQL commands

- DDL (DATA Definition language)
 - ↳ create, alter, rename, truncate & drop
- DQL (Query language) $\rightarrow$ select
- DML (Manipulation language) $\rightarrow$ insert, update & delete
- DCL (Data control language) : grant & revoke

TCL (Transaction Control Language)

start transaction, commit, rollback

CREATE DATABASE IF NOT EXISTS db-name;

DROP DATABASE IF EXISTS db-name;

SHOW DATABASES → give all databases.

SHOW TABLES;

INSERT INTO TABLE-NAME (col1, col2)
VALUES (col1, col2);

(101, HANU)

(102, HARSHIT)

(103, Kehn);

Primary Key $\rightarrow$ unique columns
• NOT NULL

Foreign Key $\rightarrow$ Primary Key of another Table

id	name	city	city
101	K	1	Pune

city id	city
1	Pune

PK

• can have dupli & null

CREATE TABLE temp (

cust id int,
FOREIGN KEY (cust id) references
customer (id)

);

constraints

- NOT NULL
- UNIQUE

col1 int NOT NULL
col2 int UNIQUE

```
# CREATE TABLE emp (
  id INT
  name VARCHAR(50)
```

PRIMARY KEY (id, name);

Default
Salary INT DEFAULT 25000

Check
Constraint age CHECK (age >= 18 AND city = "CHD");

CHECK
age INT CHECK (age >= 18)

clause — where

SELECT * FROM S WHERE mark > 80

- AND — ~~A AND B~~ Both True A AND B
- OR → one true A OR B
- BETWEEN → give range A BETWEEN B
(range)
- In → match list

SELECT * FROM Student WHERE CITY IN ("Delhi", "Mumbai");

- NOT IN → NOT match list

• LIMIT

SELECT * FROM Student LIMIT 3;

Order By

SELECT * FROM STA

ORDER BY city ASC;

↳ ascending

ORDER BY city DESC;

Aggregate functions

COUNT()

MAX()

MIN()

SUM()

AVG()

SELECT mode, count(custom)
FROM payment
GROUP BY mode

GROUP BY

SELECT city, count(name)

FROM student

GROUP BY city;

HAVING → ~~conditions~~ groups

WHERE → ~~conditions~~ clause

SELECT Column
FROM Table name
WHERE condition
GROUP BY column
HAVING condition
ORDER BY column ASC;

where applies condition on (rows)
having → groups

where
GROUP BY
HAVING

TABLE OPERATION

classmate

Date

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UPDATE

```
UPDATE TABLE_Name  
SET col1 = val1, col2 = val2  
WHERE condition;
```

```
UPDATE STA  
SET grade = 'O'  
WHERE grade = 'A';
```

(modify & update)

Safe mode

```
SET SQL_SAFE_UPDATES = 0;
```

Delete

```
DELETE FROM TB_Name  
where marks < 33;
```

cascading in FK

- on DELETE CASCADE
 - on UPDATE CASCADE
- Auto Adjust
Determin FK

TABLE - ALTER

(To change schema)

- ADD column

```
ALTER TABLE TB_name  
ADD COLUMN col_name col_type constrain
```

- Drop column

```
ALTER TABLE TB_name  
DROP COLUMN col_name
```

- Rename Table

```
ALTER TABLE TB_name  
RENAME TO new - Table_name;
```

o change column (rename)

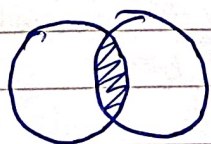
ALTER TABLE Table_name
change column old_name new_name new_datatype
new_constraint;

o MODIFY column (datatype / constraint)

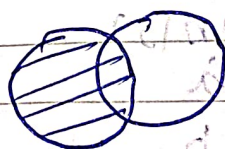
ALTER TABLE Table_name
MODIFY col_name new_datatype new_constraint

TRUNCATE — delete Table DATA
TRUNCATE TABLE STUDENT;

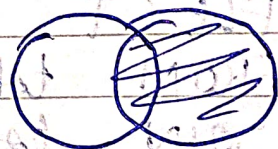
JOIN — combine rows



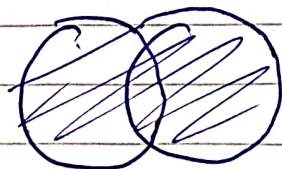
inner JOIN



left JOIN



right JOIN



full JOIN

Outer JOIN

Inner JOIN (inner) matches records

SELECT column(s) FROM TableA
INNER JOIN TableB
ON TableA.col_name = TableB.col_name;

full JOIN

LEFT JOIN — ko cam

RIGHT JOIN — ko cam

RIGHT JOIN — ko cam

self JOIN

SELECT column(s)
FROM table as a
JOIN table as b
ON a.col_name = b.col_name;

self JOIN

self JOIN

SQL Subquery

SELECT column
FROM table_name

WHERE column_name operator
(subquery);

VIEWS

CREATE VIEW view1 AS

SELECT rollno, name FROM STUDENT

SELECT FROM view1;

VIEW NAME

Derived tables are temporary tables created from the result of a query.

EX -

SQL

SELECT

VIEW